

MOHD ANAS

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EDUCATION

B.Tech in Information Technology

August 2023 - May 2027

IIIT Vadodara, Gujarat, India

Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Design, Backend Systems

EXPERIENCE

Aarogya Kaya | Remote | Software Engineer Intern

July 2025 - January 2026

- **Architected secure backend services** using CQRS, PostgreSQL, and Supabase RPCs, protecting **15 RPC functions and database tables** with **JWT/RBAC, Row-Level Security, admin-only controls, environment secrets, and least-privilege access**.
- Optimized **backend read paths** with in-memory caching, Redis, and database access patterns, reducing average latency from **420ms to 38ms** while maintaining **p95 latency below 250ms**.
- Engineered **fault-tolerant asynchronous pipelines** with retries and idempotent processing for a production service serving **1,000+ users daily**.
- Established **CI/CD pipelines** for backend and web services, automating **unit tests, Playwright E2E tests, linting, build validation, and deployment** across pull requests and merges.

Open Source | Remote | Software Contributor

July 2025 - May 2026

- Migrated the **Next.js (Vercel) patch-next utility** from CommonJS to TypeScript ESM, saving **15–30 seconds** per local reinstall during development and testing.
- Delivered **22 merged pull requests and 12 issues** through maintainer reviews, **design discussions, and iterative feedback**, translating reported problems into fixes and features while improving developer tooling.

PROJECTS

Second Brain – Agentic Decision Simulator | Python, LLMs, Multi-Agent Systems, RAG

[Live](#) | [Github](#)

- Built a **multi-agent decision system** with **3 specialized agents** across a **4-stage, 2-pass workflow**, orchestrating retrieval, scenario generation, and what-if expansion with **2 LLM calls per initial run** and **2–6 generated nodes** per expansion.
- **Raised retrieval hit rate from 75% to 90% and groundedness to 92%** across **10K chunks** using query rewriting, semantic retrieval, and reranking; added an evaluation/guardrail harness with **12 validation checks** that caught **95% of invalid graphs** across 3 failure categories.

A3.store – Distributed Key-Value Database | Python, Azure AKS, AWS EKS, Kubernetes, Terraform, gRPC

[Live](#) | [Github](#)

- Developed a **leaderless distributed key-value database** with **gRPC replication, quorum consistency, gossip-based failure detection, hinted handoff, and anti-entropy repair** across a **3-node cluster**.
- Managed **stateful Kubernetes workloads with Terraform across Azure AKS and AWS EKS**, running 3 KV replicas with autoscaling node pools, pod anti-affinity, persistent volumes, rolling updates, PDBs, and health probes; instrumented **10 Prometheus metrics** with Grafana for production troubleshooting.

TECHNICAL SKILLS

Languages: C/C++, Go, Python, JavaScript, SQL

AI & Agentic Systems: LLMs, Multi-Agent Systems, RAG, Semantic Retrieval, Reranking, Evaluation, Guardrails

Azure & Backend: Microsoft Azure, AKS, Azure SQL, Azure Blob Storage, Azure Monitor/Application Insights, REST APIs

Cloud, DevOps & Quality: GitHub, GitHub Actions, Kubernetes, Docker, Terraform, CI/CD, PostgreSQL, Redis, Prometheus, Grafana, Pytest, Playwright

ACHIEVEMENTS

- Ranked **365th** among **15,804** participants in CodeChef **START192D**.
- Ranked **3,026th** among **17,000+** participants in Codeforces Round 1110 (Div. 1 + Div. 2).

LEADERSHIP & ACTIVITIES

- Organized an intra-college hackathon and mentored **25+ teams** as a **Technical Committee Core Member**.
- Led recruitment, mentorship, and event initiatives for **100+ students** as **Joint Secretary of DOT Club**.